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The Prevalence of High School Multi-Sport Participation in Elite National Football League Athletes

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The Prevalence of High School Multi-Sport Participation in Elite National Football League Athletes

ABSTRACT

Objectives: Early youth specialization is increasingly being researched and identified as a risk factor for overuse injuries and burnout. This study aimed to characterize high school sport specialization in top National Football League (NFL) athletes and determine whether associations exist between single-sport specialization and future injury risk, performance, and longevity.

Methods: For NFL first-round draft picks from 2008-2017, data on the number of games played in the NFL, overuse injuries causing athletes to miss one or more regular season games, Pro Bowl selections, and current status in the NFL were collected using publicly available information.

Results: A total of 318 athletes were analyzed. Multi-sport athletes were highly prevalent (88%, n=280), while only 12% (n=38) of athletes were classified as single-sport. No difference between multi-sport and single-sport athletes was found regarding games missed to upper (p=0.93) or lower extremity injuries (p=0.49), total games played

(p=0.57), or NFL longevity (p=0.97). There was no significant difference in the proportions of athletes reaching at least 1 Pro Bowl.

Conclusion: The majority of NFL first-round draft picks were multi-sport athletes in high school. Single-sport football participation in high school does not appear to aid athletes in reaching or succeeding in the NFL.

Keywords: sport specialization; athletic injuries; youth sports; performance

INTRODUCTION

Participation in youth sports is widespread and continues to grow across the United States [1]. A concomitant rise in youth sporting injuries has been observed in these young athletes, with early sports specialization as a potential cause [2,3]. Definitions of "early sport specialization" vary, but it generally refers to training and/or competing in a single sport for most of the year, to the exclusion of other sports, by around age 12 [4]. Here, we define early specialization as participating in a single varsity sport in high school, playing no other varsity sports between 9th and 12th grade. The sports medicine community has increasingly linked early specialization to burnout and risk of overuse injuries [4-7]. While the etiology of sport specialization is likely multifactorial, exogenous influences from parents, coaches, and trainers have all been identified as contributing [8-10].

The American Orthopaedic Association for Sports Medicine, The American Academy of Pediatrics, and the National Athletic Trainers' Association are among many national organizations that have recommended children and adolescents participate in multiple sports and that they take 2-3 nonconsecutive months off from a specific sport per year [11,12]. These organizations have strongly advocated for multi-sport participation in

youth to decrease injury risk, decrease burnout, and augment overall athletic dexterity [10]. Despite recommendations from these organizations, however, young athletes continue to sustain pressure to specialize even from their parents and coaches [8,13].

Brooks et al recently demonstrated that highly specialized young athletes were significantly more likely to believe they would receive a college scholarship than their multi-sport peers [14]. In another study of athletes between 7 and 18-years-old (n=235), approximately one-third of young athletes reported being told by a coach to specialize in a single sport, and 60% were playing a primary sport for 9 or more months per year [9]. This study found that playing a primary sport for a greater proportion of the year was correlated with a history of injury [9], and there is additional evidence that athletes specializing at younger ages are more likely to suffer overuse injuries [15-18]. Given these potential risks, it is important to determine whether there are benefits to early sport specialization.

While there is significant interest in the effects of sport specialization on young athletes, there is limited research on the impact of sport specialization upon elite athletes. Rugg et al investigated high school specialization in National Basketball Association (NBA) athletes, linking early specialization to injury risk and shorter career longevity [19]. Similarly, in Major League Baseball, early specialization was linked to playing fewer games and a higher incidence of upper extremity injuries [20]. Additionally, multiple studies have shown increased success in elite and even Olympic athletes with multi-sport engagement during youth [21,22].

The link between youth specialization and future injury risk has recently been investigated in the NBA and MLB, but no such study has been conducted to date in

professional football players. The purpose of this study was to (1) describe the prevalence of high school sport specialization in National Football League (NFL) first-round draft picks, (2) assess respective injury rates in multi-sport athletes versus single-sport athletes, and (3) examine professional career longevity between multi-sport athletes and single-sport athletes. We hypothesized that NFL players that played in multiple sports during high school would sustain a lower rate of injuries and outperform those who specialized in football in terms of Pro-Bowl selection and longevity in the league.

MATERIALS AND METHODS

First-round draft picks from the ten-year span of 2008 to 2017 were included in this study. First round draft picks represented a higher-level football player and were likely to have more public information available as well as a greater likelihood of playing professionally. An athlete was considered a multi-sport athlete if there was evidence that they played another sport in high school based on publicly available information on the internet from trackingfootball.com, hudl.com, maxpreps.com, athletic.net, and other resources such as local newspaper profiles and college football athlete biographies. Given the limited publicly available data on pre-high school activities, sports participation in middle school and earlier was not evaluated. If there were no indications that an athlete played another sport in high school they were classified as a specialist.

Draft data (height, weight, age) were gathered from nflcombineresults.com, and body mass index (BMI) was calculated. Players were classified into one of two position categories. “Agility” players included running backs, wide receivers, tight ends, quarterbacks, cornerbacks, safeties, and linebackers. “Linemen” included centers, offensive

guards, offensive and defensive tackles, and defensive ends. Current active versus inactive status in the NFL was determined by searching the current player database on nfl.com.

The number of regular-season games played in the NFL and Pro-Bowl appearances were gathered from pro-football-reference.com. Preseason and playoff games were excluded. Off-season injuries were also excluded because of limited publicly available information. Data on injuries sustained in the NFL and the number of games missed as a result was gathered from espn.com, nfl.com foxsports.com, sportsinjurypredictor.com, rotoworld.com, and other online news sources. The location and all publicly available details about the injury were recorded. Injuries that did not cause players to miss any regular season games were excluded. Injuries sustained outside of the NFL as well as traumatic impact injuries including concussions, brachial plexus injuries or "stingers," and any fractures, were excluded. Unspecified hand and foot injuries were also excluded due to their frequent traumatic origin. Injuries were classified as upper extremity (wrist, arm, shoulder, and pectoral/chest injuries), lower extremity (ankle, lower leg, knee, thigh, groin, and hip injuries), or core (abdominal or back injuries).

Games missed due to non-injury reasons such as suspension or illness were excluded from the total number of games. Instead, the possible number of games was obtained by tabulating total games and subtracting games missed for non-injury reasons [19]. Concordantly, the percentage of games missed due to a certain type of injury was calculated using possible games as the denominator.

Data were compiled and analysis were completed using student's t-tests for continuous variables including age, height, weight, BMI, and percentage of possible games played. Chi-square tests were used for categorical variables including "agility" vs. "lineman"

classification, pro-bowl selection, and active vs. inactive NFL status. Statistical significance was defined as $p < 0.05$.

RESULTS

A total of 318 athletes were drafted to the NFL in the first round from 2008 to 2017. The prevalence of multi-sport participation was 88.1% (n=280), whereas 11.9% (n=38) of this cohort was classified as a single-sport, football-specialized athlete. The most common additional high school sport was track (n=196), followed by basketball (n=170), baseball (n=38), wrestling (n=10), lacrosse (n=3), golf (n=3), tennis (n=2), soccer (n=2), hockey (n=1), and rugby (n=1). Knee, ankle, hamstring and shoulder injuries were the four most likely injuries to cause NFL players to miss one or multiple games.

The age, height, weight, and BMI of single- vs multi-sport athletes are shown in **Table 1**, and the proportions of each cohort that was drafted into agility vs. linemen positions are shown in **Table 2**. While single-sport athletes weighed 15.3 lb more (p=0.052) than multi-sport athletes and maintained a corresponding mean BMI 1.4 higher (p=0.084), these findings were not statistically significant.

The injuries causing athletes to miss regular season games are shown in (**Table 3)**. Only the location of the injury, rather than the subtype and nature, were publicly available for the majority of injuries. When comparing multi-sport (n=280) to single-sport (n=38) athletes, there were no significant differences in the mean percentage of total games played (88.1% ± 1.4% vs. 86.7% ± 1.4%, p=0.571), the mean percentage of games missed due to upper extremity injuries (2.2% ± 5.9% vs. 2.3% ± 7.0%, p=0.928), or the percentage of games missed due to lower extremity injuries (9.1% ± 1.2% vs. 10.5% ± 1.1%, p=0.485).

While 34.6% of multi-sport athletes reached the Pro Bowl, only 28.9% of single-sport athletes reached the Pro Bowl. This finding was not statistically significant ($p = 0.49$). These findings are tabulated in Table 4. Regarding longevity, 73.9% of multi-sport athletes and 73.7% of single sport athletes studied were still active in the NFL. This finding was not statistically significant ($p=0.097$) and is summarized in Table 5.

DISCUSSION

The purpose of this study was to characterize sport specialization in first round NFL draft picks and determine whether associations exist between playing multiple sports in high school and future injuries, performance, and longevity in the NFL. Fewer than 12% of the athletes reviewed in this study specialized only in football while competing at the high school level with a large proportion of them having participated in track and/or basketball. No significant differences in the percentage of possible games played, Pro Bowl selection, or likelihood of still being active in the NFL today was found between single-sport and multi-sport athletes.

While prior studies have assessed youth specialization with Olympian, NBA and MLB athletes this study is the first to analyze youth specialization in elite NFL players [19-22]. In contrast to NBA and MLB players, our study demonstrated an extremely low rate of single-sport high school specialization, which could be demonstrative of the balanced athletic dexterity required in professional football. Conversely, this difference may be attributable the fact that football in many areas is a single-season sport, whereas other

sports often have two seasons or athletes playing on travel teams in addition to a school team. In the same vein, playing only football in high school does not automatically translate to playing football year-round.

While a higher career longevity and lower injury risk were demonstrated in NBA and MLB multi-sport athletes, no such link was found for NFL players. This can likely be explained by the traumatic nature of NFL injuries, which has been demonstrated in multiple epidemiological investigations of NFL shoulder and upper extremity injuries [23-25]. While fractures and dislocations were the two most common injuries in professional football overuse ligamentous injuries were the most common upper extremity injuries in baseball players [26,27].

This study indicates that the majority of first round NFL players were multi-sport athletes in high school. The skills and dexterity gained playing additional sports may actually help players continue to advance to the highest level, a theory popularized in studies of other elite athletes [21,22]. Additionally, multi-sport players demonstrate significantly better neuromuscular control than single-sport players when tested with a jump-landing test [28]. Of course, it is also possible that sport specialization neither helps nor hinders advancement to elite play and longevity in the NFL.

Youth athlete specialization is likely a significant contributor to the increase in overuse injuries found in adolescent athletes, though this is beyond the scope of this study. Despite staunch opposition from national organizations such as the American Orthopedic Society for Sports Medicine and the American Academy of Pediatrics players continue to specialize in a single sport a young age [8,13]. The AOSSM staunchly opposed year-round participation and exclusively playing a single sport prior to age 12 [4]. In spite of this,

demographic studies have found specialization often begins even before the age of 10 and that exogenous pressure from parents and coaches is prevalent [8,16,17]. Investigations such as these are crucial to debunking erroneous, popularized beliefs that young children must engage in single-sport participation to maximize their potential for collegiate and professional play. Future prospective studies following young athletes into adulthood can help elucidate the relationship between playing multiple sports, athletic dexterity, injury risk, and eventual professional play.

Several limitations were present in this investigation. Only 38 athletes were classified as single-sport, and this sample size may have been underpowered to identify differences between single and multi-sport athletes. Additionally, using information from publicly available sources on the internet may have led to the misclassification of athletes as multi- vs. single-sport or inaccurate recording of games missed due to injuries. It is possible that players took part in other sports in high school at a less-competitive level which did not show up in our searches. An additional limitation was in injury detail as NFL injury reports list the body part affected but no further details. Finally, we excluded off-season injuries because of limited information, which could have had an impact on our results.

CONCLUSION

The vast majority of first round NFL players were multi-sport athletes in high school. No significant differences in the rate of injuries, longevity in the league, or elite performance level were found between single-sport versus multi-sport NFL players who were selected in the first round of the draft between 2009 – 2017. These findings support

the value of youth multi-sport participation and suggest that there may not be long-term performance benefits to early sport specialization.

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Table 1. Age, Height, Weight and BMI in Single-sport vs. Multi-sport Athletes

	Age (y)	Height (in)	Weight (lb)	BMI
Multi-sport	22.0 ± 1.0	74.5 ± 2.5	250.5 ± 45.6	31.6 ± 4.4
Single-sport	22.3 ± 1.1	75.1 ± 2.0	265.8 ± 44.2	33.0 ± 4.5
P-value	0.078	0.061	0.052	0.084

Table 2. Type of Position in Single-sport vs. Multi-sport Athletes

	Agility % (n)	Lineman % (n)
Multi-sport	58.9 (165)	41.1 (115)
Single-sport	44.7 (17)	55.3 (21)
P-value	0.097	

Table 3. Injuries Causing Athletes to Miss ≥ 1 Regular Season Games

Injury Location	No. Reported
Knee (no ACL involvement)	133
Ankle	124
Hamstring	97
Shoulder	53
ACL	41
Hip or Groin	27
Back	27
Calf	16
Pectoral or Chest	15
Biceps	11
Achilles	10
Quadriceps	9
Elbow	6
Triceps	6
Wrist	3
Plantar fascia	3
Abdomen	2
Thigh	2
Shin	1

Table 4. Selection to Pro Bowls in Single-sport vs. Multi-sport Athletes

	Made ≥ 1 Pro Bowl	No Pro Bowl Selections
	% (n)	% (n)
Multi-sport	34.6 (97)	65.4 (183)
Single-sport	28.9 (11)	71.1 (27)
P-value	0.487	

Table 5. Current Status in the NFL in Single-sport vs. Multi-sport Athletes

	Active % (n)		Inactive % (n)
Multi-sport	73.9 (207)		26.1 (73)
Single-sport	73.7 (28)		26.3 (10)
P-value		0.974	